

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and captured schematics for **optical control circuits** and **signal chain processing**, focusing on components suitable for space-grade hardware.
- Developed and simulated **Printed Circuit Board (PCB) layouts** for next-generation Free Space Optics (FSO) transceivers, including exposure to high-speed RF layout principles.
- Created and specified **power budgets** for subsystems, ensuring adherence to AS/EN9100 and relevant JEDEC standards for aerospace deployment.
- Modeled and simulated **analogue readout circuits**, including TIAs and DACs, validating performance through SPICE simulations for aerospace applications.
- Contributed to the development of subsystems from prototype to aerospace-qualified products, working closely with interdisciplinary mechanical and optical teams.
- Documented and reviewed the design, test, and validation procedures for subsystem designs, ensuring compliance with aerospace standards.
- Containerized and deployed backend services for a **payments operations dashboard** on **AWS EC2** instances using **Docker**, managing CI/CD pipelines with **Jenkins** for a team of 8 engineers.

SKILLS

Electronics Design – Analogue Circuit Design, Digital Circuit Design, Schematic Capture, PCB Layout, Signal Integrity, Impedance Matching, High Speed RF Layout, Power Budgets

Simulation & Modelling – SPICE Simulation, Noise Models, Link Budgets, Thermal Modelling

Aerospace Standards – AS/EN9100, JEDEC, Space-Grade Hardware

Communications Protocols – CAN, SPI, UART, I2C

Laboratory Equipment – Oscilloscopes, Multimeters

Software Tools – Altium, EAGLE, KiCAD, PADs, RStudio, Pandas, Scikit-learn

Backend Development – Java, Spring Boot, Python, REST APIs, AWS, Docker, Jenkins, PostgreSQL

PROJECTS

Space-Grade FSO Transceiver Simulation

- Simulated analogue readout and signal processing circuits for FSO transceivers using SPICE, focusing on TIAs and DACs.
- Developed PCB layout designs incorporating high-speed RF principles and impedance matching for optimal signal integrity in aerospace applications.
- Created detailed power budgets and noise models to predict subsystem performance under aerospace environmental constraints.

Payments Operations Dashboard Monitoring Service (HCL Technologies)

- Engineered a Python-based service deployed on **AWS Lambda** to monitor transaction flows and generate alerts for failed payment messages.
- Integrated with **Splunk** for log analysis and error detection, reducing manual triage time for P1 incidents by approximately 30%.
- Developed REST APIs using **Java Spring Boot** to expose operational metrics for the financial-services client's dashboard.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)